THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** basic	turns red litmus blue
2	**(b)** acids formed from sugar	bacteria make acids; toothpaste is mildly basic
3	**(d)** the metal base and the tip	contact points
4	**(d)** current passes through wet skin	body conducts
5	**(a)** Sulfur	a yellow non-metal
6	**(c)** copper	metals conduct
7	**(d)** physical change	returns to shape
8	**(b)** evaporation	physical process
9	**(d)** puberty	onset of adolescence
10	**(d)** normal emotional changes	due to changes in the body
11	**(a)** conduction	heat flows along the rod
12	**(d)** Wood	used as handles
13	**(b)** 60 km/h	speed = distance ÷ time
14	**(b)** 60 km	distance = speed × time
15	**(a)** gills	dissolved oxygen
16	**(b)** rumination	partially digested food returns to mouth
17	**(a)** green and non-green patches	chlorophyll in patches
18	**(b)** root hairs	absorbing hairs
19	**(b)** vacuum	about 3×10^{8} m/s
20	**(b)** Moon’s umbra	narrow path

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Dip filter paper in turmeric paste, dry and cut into strips (2).	2
22	Cell and battery; mains supply (2).	2
23	It reacts vigorously with air and water (2).	2
24	Rapid changes in feelings due to hormonal and emotional changes (2).	2
25	Hot air rises, warming the room by convection (2).	2
26	Motion along a straight line (1); a train on a straight track (1).	1+1
27	Glucose + Oxygen → Carbon dioxide + Water + Energy (2).	2
28	Absorb water and minerals from soil (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Steel spoon and pencil lead (3).	3
30	Oil floats and spreads, making the fire worse (3).	3
31	Convection currents and radiation (3).	3
32	Supply helpful bacteria (3).	3
33	Dressing, vehicles (1.5); periscopes, torches (1.5).	1½+1½
34	No seasons (1.5); equal days and nights (1.5).	1½+1½
35	A neutral solution does not taste sour (1); does not change the colour of litmus (1); an acidic solution tastes sour and turns blue litmus red (1).	1+1+1
	OR: Do not taste (1); wear gloves and goggles (1); handle with care (1).	1+1+1

36	Make own food (1.5); food source for all consumers (1.5).	1½+1½
	OR: Chlorophyll (1.5); larger absorbing surface (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Paint wears off (1). (ii) Rust weakens it (1). (iii) Galvanisation (1). (iv) Stainless steel (2).	1+1+1+2
	OR: (i) Three set-ups (2). (ii) Rust only in air plus water (2). (iii) Both needed (1).	2+2+1
38	(i) Hoarse voice (1). (ii) Voice box (1). (iii) Secondary sexual characteristic (1). (iv) No, timing varies (2).	1+1+1+2
	OR: (i) Definitions (2). (ii) Height, voice, hair, acne, body shape (2). (iii) Individual pace (1).	2+2+1
39	(i) 180 km (1). (ii) 3 h (1). (iii) 60 km/h (1). (iv) No, unequal distances (2).	1+1+1+2
	OR: (i) Distance/time; m/s (1). (ii) 60 km/h (2). (iii) 16.67 m/s (2).	1+2+2
40	(i) Transparent, translucent, opaque (3). (ii) Opaque (1). (iii) Block all light (1).	3+1+1
	OR: (i) i = r (1). (ii) Beam on mirror, measure (2). (iii) From smooth surface (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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